

SSN College of Engineering, Kalavakkam
Department of Computer Science and Engineering
IV Semester – B.E (CSE)
R2024
UCS3461 - Foundations of Artificial Intelligence (TCP-Lab)

Academic Year: 2025-2026 Even

Batch: 2024-2028

Ex: 9 Intelligent Multi-Device Energy Management System using Classical Planning and Decision Networks

(CO5, CO6, K4, PIs: 2.1.3, 3.1.1, 3.2.1, 3.4.2, 4.1.2, 4.2.1, 5.1.1, 11.2.1, 12.3.2, 12.4.1, 13.4.1)

Design and implement an Intelligent Energy Management System for a smart environment (such as a hospital ICU, smart home, or electric vehicle charging station) using **classical planning (forward state-space search) and decision networks**. The system must manage the charging of multiple devices (3–5 devices) with different priorities, battery levels, and usage patterns, while operating under limited charging resources (e.g., only two or three devices can be charged at a time). Each state represents the battery levels of all devices, i.e., $S = (B_1, B_2, B_3, \dots, B_n)$ and actions correspond to selecting a subset of devices to charge at each time step. Battery levels increase for charging devices and decrease for others based on their consumption rates, while ensuring values remain within valid limits (0–100). Starting from an initial state, generate all possible charging schedules over a fixed planning horizon (e.g., 5–8 steps) by exploring different combinations of actions using forward search, where each action leads to a new state based on these transition rules.

To model real-world uncertainty, incorporate a decision network framework where charging efficiency, device usage, and unexpected power variations are probabilistic. Assume charging succeeds with probability p (e.g., 0.85–0.95) and failure occurs with probability $1 - p$. Define a utility function that reflects system priorities: assign higher rewards for maintaining high-priority devices above safe battery thresholds, moderate rewards for normal devices, and penalties for undesirable conditions such as low battery levels, battery depletion, and frequent switching of charging decisions. For each generated plan, compute the Total Cost and Expected Utility (EU) using the following formulas:

$$\text{Battery Deficiency Cost at step } t = \sum_{i=1}^n (100 - B_i^t)$$

$$\text{Total Cost} = \sum_{t=1}^H \left[\sum_{i=1}^n (100 - B_i^t) \right] + (\text{Number of Switching Actions} \times C_s)$$

(where C_s is the switching penalty, e.g., 5)

$$P(\text{success}) = p^H, P(\text{failure}) = 1 - P(\text{success})$$

$$EU = P(\text{success}) \times \text{Total Reward} - P(\text{failure}) \times \text{Total Penalty} - \text{Total Cost}$$

The system must evaluate all possible charging plans and identify the optimal schedule that maximizes expected utility while ensuring reliability and safety of critical devices.

The system should also include comprehensive visualization and analysis tools. Students must visualize how battery levels of all devices evolve over time and clearly highlight the selected optimal charging schedule. The state-space search process should be represented as a tree (for a smaller horizon such as 2–3 steps for clarity), where nodes represent battery states and edges represent charging actions. Analytical visualizations such as expected utility comparison, device-wise battery stability, and failure risk analysis should be included.

Note: Use generative AI tools to explore the concepts and present your work in a colorful and impressive way.

Evaluation Rubric:

S.No	Component	Description	PIs	Marks
1	Problem Definition & Modeling	Clearly defines state, actions, transitions, objectives, constraints, and assumptions	3.1.1, 2.1.3	10
2	Design of Planning Approach	Proper implementation of forward state-space search, planning horizon, and generation of all plans	3.2.1, 4.2.1	15
3	Decision Network & Utility Design	Correct modeling of uncertainty, probabilities, reward/penalty structure, and Expected Utility calculation	2.1.3, 4.1.2	15
4	Implementation & Integration	Functional code integrating state transitions, planning, and decision-making modules	3.4.2, 12.3.2	15

S.No	Component	Description	PIs	Marks
5	Testing & Experimentation	Use of multiple test cases, varying initial states, and meaningful experimentation	4.1.2, 12.3.1	10
6	Visualization & Presentation	Graphs (battery vs time, EU comparison), optional state-space tree, clarity of outputs	4.2.1	10
7	Analysis & Interpretation	Clear explanation of results, justification of optimal plan, insights and observations	12.4.1	10
8	Report Quality & Documentation	Well-structured report including problem description, methodology, results, graphs, and conclusions	12.3.2, 12.4.1	10
9	Tools, Innovation & Future Scope	Use of tools (Python/libraries), discussion of real-world relevance, improvements, emerging technologies	5.1.1, 11.2.1, 13.4.1	5